American International University-Bangladesh (AIUB)

Department of Computer Science Faculty of Science & Technology (FST)

PROJECT PROPOSAL

A Software Engineering Project Submitted
By

Semester : Spring: 25-26		**Section: F**
SL	**Student Name**	**Student ID**
1	**MD.LEON MIA**	**23-53551-3**
2	MD. ABUBOKOR SIDDIK ROJAN	22-47890-2
3	FARHAN KABIR	23-53686-3
4	MONIA ISLAM MUKTI	24-55933-1

1.Project Title - SkillSwap: A Community-Driven Peer-to-Peer Learning and Skill Exchange Ecosystem.

2. Introduction :

Personal growth and empowerment are essential in today's evolving world, making the acquisition of new skills vital. However, many individuals struggle as they depend on informal or unstructured methods to learn or share expertise. This project introduces SkillSwap, a digital system designed to facilitate direct connection between those who have valuable knowledge and those seeking to learn, fostering community collaboration without financial barriers.

3. Background to the Problem

Many people have valuable skills but cannot find the right mentors or students due to the lack of a structured system. The main weaknesses of current methods are:
Difficulty in Discovery: There is no central way to know who is skilled in what within a local area.
Financial Barriers: High-quality courses are often too expensive for many people.
Lack of Structure: It is difficult to maintain time and quality when learning through informal, unorganized ways.

4. Problem Statement

There is no organized digital system for skill exchange. As a result, experts lack a proper environment to share their knowledge, and eager learners fail to develop their talents due to a lack of mentors. A specific, technology-driven platform is needed to bridge this gap.

5. Objectives of the Project

- To build a user-friendly web interface where users can easily find mentors or students.
- To create a learning culture through direct skill exchange without any monetary transactions.
- To build a strong network and cooperation among skilled local individuals.

6. Proposed Solution Overview

We propose a web-based digital system that works on a Peer-to-Peer (P2P) model.
Technology Stack: The system will use modern frameworks and databases to ensure fast response times and data security.

How it Works: Users can create skill profiles and send learning requests after viewing others' profiles. A matching algorithm will connect the right mentors with the right students.

7. Target Users

Students: Those who want to gain new academic or technical skills.

Job Seekers: Those interested in learning specific skills required for the workplace.

Skilled Individuals: Experts who want to share their knowledge with others

8. Scope of the Project

Included: User registration, login system, profile management, and skill searching features.

Limitations: This will be a web platform developed within a single semester to show end-to-end functionality. Payment gateways and mobile apps are currently out of scope.

9. Expected Impact and Benefits

- Users can learn new skills without any financial cost.
- Social interaction and mutual trust within the community will increase.
- It will create a knowledge-sharing ecosystem that helps in long-term education and job opportunities.

10. Conclusion

SkillSwap will remove existing barriers to skill exchange by providing a centralized platform. This is not just an academic project; it is a practical step toward solving real-world problems and empowering the community. By combining academic knowledge with real-world application, this project will prove to be a successful solution.